Tiffany Yu

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EDUCATION

University of South Carolina *Columbia, SC*
Bachelor of Science in Computer Engineering **Expected Graduation: December 2026**
GPA: 3.483

SKILLS

Technical: Programming Languages | Robotic systems and tools | Communication Protocol – UDP,TCP | Network edges | AutoCAD | Graphic Design | Spice | LabVIEW
Operating Systems: Window 10&11 | Ubuntu | Linux | Android
Programming: Python, Java, Git Repository, C++, Docker, ROS, MATLAB, LaTeX, Assembly Language
Languages: Fluent in Chinese, Beginner in Spanish

WORK EXPERIENCE

University of South Carolina *Columbia, SC*
Discrete Structures Math Grader **Dates: August 2025 – Current**

- Graded assignments, homework, and quizzes, in Discrete Structures, ensuring consistency, accuracy, and timely feedback for students.
- Provided written feedback that helped students improve reasoning, proof structure, and conceptual understanding.
- Managed grading workload under deadline pressure, ensuring high quality and fairness while maintaining confidentiality.

Undergraduate Research Assistant **Dates: January 2026 – Current**

- Designed and implemented a multi-layer perceptron on an NI cRIO-9033 FPGA target using LabVIEW.
- Built a pipelined matrix multiplication engine using a XNode state machine within a Single-Cycle Timed Loop (SCTL), managing pipeline latency and DMA FIFO data transfer.
- Managed time effectively to complete the project on schedule despite a short timeline, balancing technical challenges and deadline pressure while maintaining quality of work.

Undergraduate Research Assistant **Dates: December 2024 – January 2026**

- Conducted research on network edge structures and cost optimization, analyzing performance trade-offs in hardware and software implementations.
- Developed and implemented a Random Forest algorithm to prune random graphs of nodes and edges, improving computational efficiency and scalability.
- Collaborated on coding a software solution to simulate and optimize network architectures, leveraging advanced deep machine learning techniques.

Undergraduate Research Assistant **Dates: March 2024 – January 2025**

- Assisted with setting up and maintaining underwater robotic equipment, ensuring proper function during research experiments.
- Help organize and prepare lab space and equipment before experiments, improving workflow efficiency for the team.

RESEARCH POSTER PRESENTATIONS

UOFSC Summer Symposium **Dates: July 23, 2026**
"Performance and Resource Analysis of MLP Forward Pass on LabVIEW FPGA"

UOFSC Discover Day **Dates: April 24, 2026**
SuperComputing International Conference 2026 **Dates: November 18, 2025**
"AutoSlim: Intelligent Automata Graph Optimization for Efficient Acceleration"

UOFSC Summer Symposium **Dates: July 31, 2025**
"ML-Based Automata Simplification for Symbolic Accelerators"

UOFSC Discover Day **Dates: April 25, 2025**
CSE Graduate Research Symposium **Dates: March 7, 2025**
"Enhancing Security and Efficiency in Computational Hardware Through Intelligent Optimization"

RESEARCH PAPERS

Roberts, T., Satme, J. N., **Yu, T**., Downey, A. R. J., Karakchi, R. E., Bakos, J. D. (2026). *High-Rate State Estimation of a Printed Circuit Board Under Shock Using an FPGA-Deployed Neural Network.* Proceedings of the ASME 2026 Conference on Smart Materials, Adaptive Structures and Intelligent Systems (SMASIS2026).

Yu, T., Stahle-Smith, R., Eswaramoorthi, D., Karakchi, R. (2025). *ML-Based Automata Simplification for Symbolic Accelerators.* arXiv preprint arXiv:2507.08751.

Karakchi, R., Stahle-Smith, R., Chinnasami, N., **Yu, T.** (2025). *Toward a Lightweight, Scalable, and Parallel Secure Encryption Engine.* arXiv preprint arXiv:2506.15070.

LEADERSHIP & INVOLVEMENT

University of South Carolina *Columbia, SC*

University 101 Peer Leader **Dates: Fall 2025, Fall 2026**

- Managed difficult conversations and provided peer support, enhancing communication skills and conflict resolution.
- Led classroom co-instruction, facilitating discussions and speaking in front of peers to guide academic success.

Vice President of Finance at Alpha Lambda Delta **Dates: Spring 2025 – Spring 2026**

- Managed financial records and budgets for the honor society, ensuring accuracy and accountability in all transactions.
- Oversaw collection and allocation of membership dues, supporting organization operations and events.
- Coordinated with campus offices and vendors to process payments and reimbursements in compliance with university policies.

Peer Mentorship Program **Dates: Fall 2024**

- Mentored first-year students by offering academic guidance and support, helping them navigate university life.
- Organized regular check-ins with mentees to track their progress, providing resources and advice tailored to their individual needs.
- Acted as a role model for mentees, sharing personal experiences to inspire confidence and success in academic pursuits.

Capstone Scholar Leadership Program **Dates: Spring 2024**

- Participated in in-depth discussions and coursework on leadership theories and practices, enhancing understanding of leadership dynamics.
- Developed and presented a group project, applying leadership strategies to successfully complete tasks within deadlines.

Awards

University of South Carolina *Columbia, SC*

Magellan Apprentice Award **Dates: Fall 2025**
Magellan Scholar Award **Dates: Fall 2025**
Capstone Fellowship Distinction **Dates: Fall 2025**
Capstone Scholar **Dates: Fall 2023 – Spring 2025**
Dean's Honor List **Dates: Fall 2024, Spring 2025**
Magellan Journey Award **Dates: Spring 2024**
President's Honor List **Dates: Fall 2023**